

AI-Assisted Brain MRI Segmentation Report

Preliminary quantitative analysis - for qualified clinical review
John

Important: This document reports automated segmentation measurements. It does not establish a diagnosis, replace a radiologist's interpretation, or provide treatment advice.

Examination and case information

Case ID	a2b09db40632
Case name	John
Status	Completed
Created	2026-08-29T08:25:49.442628+00:00
Report generated (UTC)	2026-09-09 11:06:06 UTC

Clinical information

Clinical indication	Not supplied with this local research case
Patient demographics	Not supplied
Comparison study	Not supplied

Technique

MRI modalities	T1, T1-CE (contrast), T2, FLAIR
Segmentation status	Completed
Volume shape	240 × 240 × 155 voxels

AI / segmentation summary

Model architecture	3D U-Net (4 input channels → 3 output channels)
Mode	Trained model checkpoint
Inference device	cpu
Segmentation labels	NCR/NET, ED, ET (BraTS convention), plus derived WT and TC
Software version	0.1.0

Quantitative results

Region	Voxels	Volume (cm ³)
Necrotic Core (NCR/NET)	15,450	15.45
Edema (ED)	42,745	42.74
Enhancing Tumor (ET)	21,929	21.93
Whole Tumor (WT)	80,124	80.12
Tumor Core (TC)	37,379	37.38

Automated segmentation findings

Whole tumor volume: 80.12 cm³. Tumor core (necrotic + enhancing) volume: 37.38 cm³. Enhancing tumor volume: 21.93 cm³. Peritumoral edema volume: 42.74 cm³. Measurements are derived from the predicted label map and voxel spacing; they should be checked against the source images before clinical use.

This is AI-generated quantitative output derived directly from the segmentation model's predicted labels — it is not a clinical diagnosis or radiological interpretation.

AI-assisted impression

Automated multi-class segmentation is present and quantitative region volumes are reported above. The result is preliminary and requires review by a qualified radiologist or clinician in conjunction with the full MRI examination and clinical context.

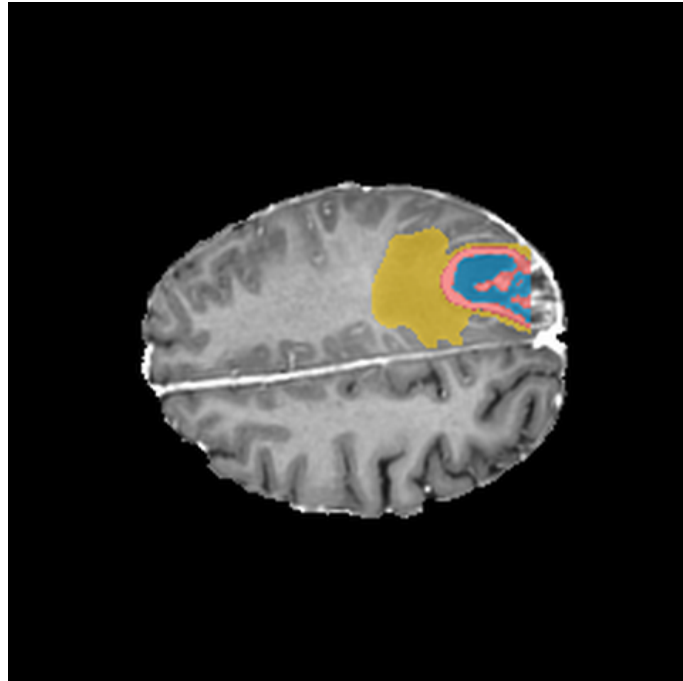
Recommended review

Review the segmentation overlay in all three orthogonal planes; verify label boundaries, image registration, and volume plausibility; and correlate with prior studies and the patient's clinical presentation when available.

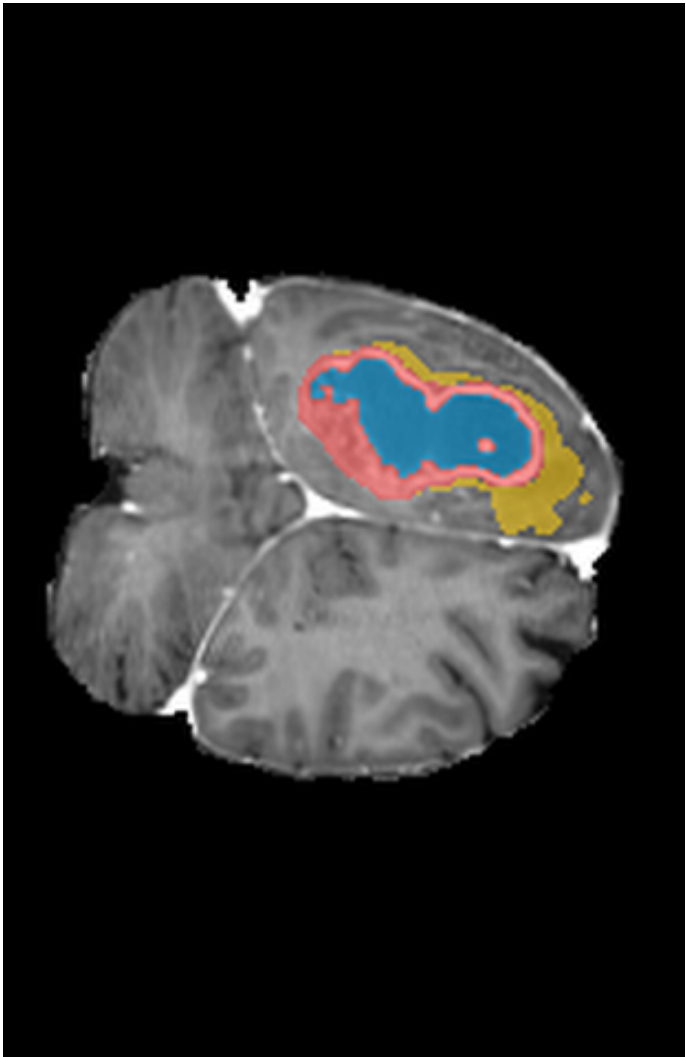
Representative visualizations

Each slice below was automatically selected as the cross-section with the largest predicted tumor area along that axis. True 3D volumetric rendering is only performed client-side in the interactive WebGL viewer and is not reproduced here.

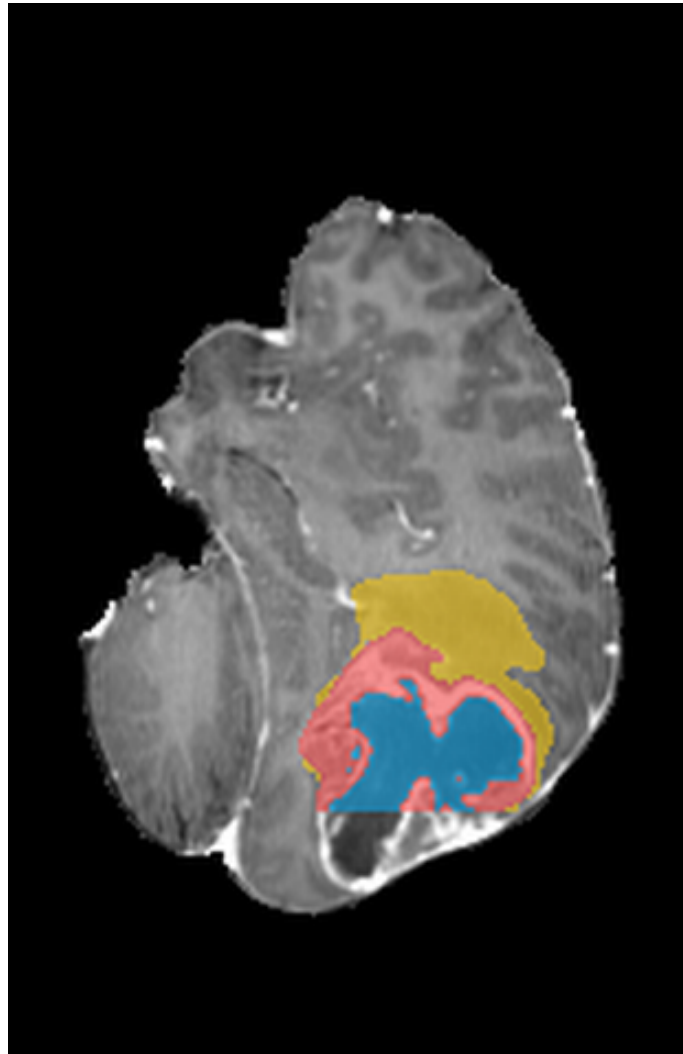
Axial



Coronal



Sagittal



Technical / model information

Checkpoint loaded	Yes
PyTorch available	Yes
Inference device	cpu

Training-time validation Dice, measured on a held-out validation set during model training — not a per-case accuracy score for this specific scan:

Region	Validation Dice
Necrotic Core (NCR/NET)	82.6%
Edema (ED)	87.7%
Enhancing Tumor (ET)	90.9%

Disclaimer

NeuroScan AI is a local, assistive research and visualization tool. The segmentation and measurements in this report are generated automatically by a machine learning model and have not been reviewed by a radiologist or clinician. This report is not a medical diagnosis and must not be used as a substitute for

professional clinical evaluation. All clinical decisions should be made by a qualified healthcare provider.